

CarbonLedger

Project Plan

Work breakdown structure, Gantt chart aligned to the contractual reporting date, and risk register

CL-PPL-001 · VERSION 1.0 · DRAFT FOR REVIEW

8

work packages

85

days of effort

CRD

aligned schedule

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Programme: B.Tech CSE 2024–28 · Semester V

Project: Emissions Accounting for a Multi-Plant Manufacturer

Project website: <https://carbon-ledger-rohan.vercel.app/>

Summary & Contents

Purpose of this document

Plans the delivery of CarbonLedger backward from the customer's contractual reporting date, with the work breakdown, schedule, effort, and risk register including supplier non-response.

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Document control

Revision history

Version	Date	Description	Author (role)
1.0	03/10/2026	Initial release for review	Sole Engineer

Approvals

Role	Signature	Date
Sponsor (Manufacturer, Head of Sustainability)		
Reporting Lead (Manufacturer)		
Sole Engineer (all project roles)		
Reviewer (Faculty)		

Related documents

CL-SRS-001 Software Requirements Specification (v1.0)

CL-DTT-001 Decision Table and Test Set (v1.0)

CL-DSG-001 Design Pack (v1.0)

CL-CMP-001 Configuration Management Plan (v1.0)

CL-DQP-001 Data Quality Plan (v1.0)

CL-PPL-001 Project Plan (v1.0) · this document

Notice. This document is controlled; printed copies are uncontrolled. Figures stated as project data (data quality tiers of $\pm 2\%$, $\pm 8\%$ and $\pm 25\%$, nine plants, three scopes) come from the case study. Numbers in worked examples and test fixtures are synthetic, are labelled as such, and are not emission factors published by any authority.

1. Project overview

1.1 Objective

Deliver CarbonLedger, an emissions accounting system for a manufacturer with **nine plants**, able to report Scope 1, 2 and 3 emissions to a customer by the **contractual reporting date (CRD)**, with every reported figure **reproducible two years later** and its uncertainty **disclosed honestly**.

1.2 Deliverables

Deliverable	Document ID	Content
SRS	CL-SRS-001	Calculation, factor versioning, uncertainty, period close and late-data rules.
Decision Table and Test Set	CL-DTT-001	48-combination table, derived tests, reproducibility regression test.
Design Pack	CL-DSG-001	Temporal data model, class diagram, reporting cycle activity diagram.
Configuration Management Plan	CL-CMP-001	Joint versioning of code and reference data (the Pair).
Data Quality Plan	CL-DQP-001	Per-plant metrics, three-cycle improvement target.
Project Plan	CL-PPL-001	This document: WBS, Gantt to the CRD, risk register.
Working system	Software	Web application, verify CLI, first issued report with golden test.

1.3 Assumptions and constraints

- **The CRD is a contract date.** Its value is not given in the case study, so this plan is expressed in **weeks counted back from it** (W0 is the week containing the CRD; W-1 is the week before). The calendar date is entered once the contract is read (task 2.1) and the whole plan shifts with it.
- **One engineer** delivers all project roles (Project Manager, Business Analyst, Architect, Developer, Tester, Configuration Manager). Capacity is taken as **5 working days a week, 20 weeks, 100 days**.
- The manufacturer provides a Sponsor, a Reporting Lead, a Reference Data Steward, and a Plant Manager and Plant Data Officer for each plant, for the reviews and data tasks in this plan.
- Technology is fixed by the SRS (Python 3.12, FastAPI, PostgreSQL 16, Git, Docker); no new tool is learned except PostgreSQL range types and exclusion constraints (risk R-17).
- Planning durations in this document (effort days, week windows, T-day calendar) are **planning estimates by the Sole Engineer**, not facts from the case study; they are revised weekly (section 8).

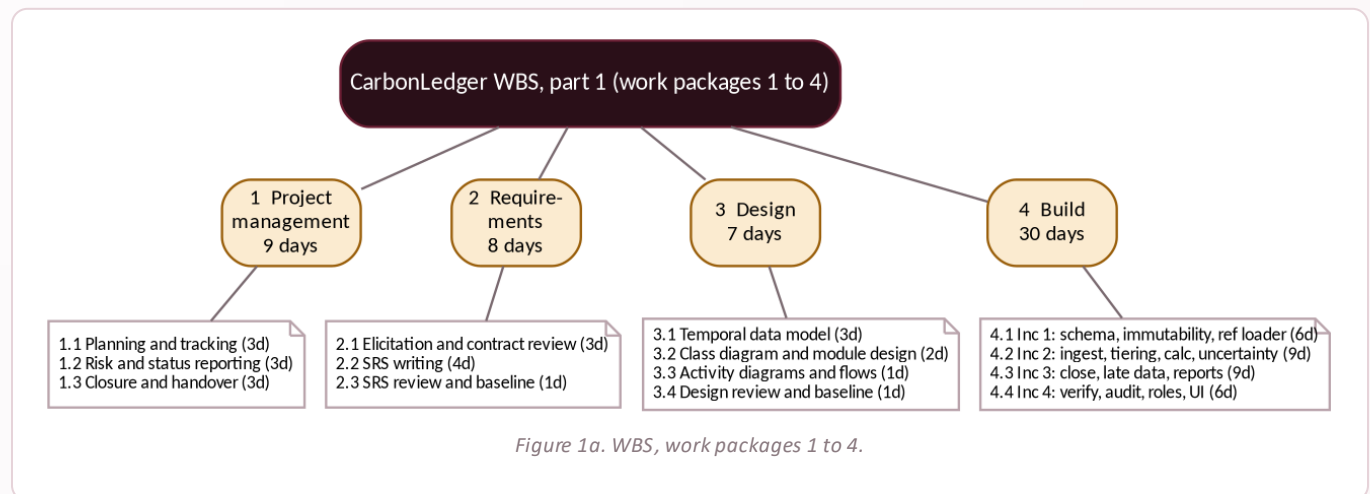
2. Approach

Lifecycle: requirements and design up front, then four timeboxed increments. Reasons: (1) auditability and reproducibility requirements must be fixed before code, because they shape the data model; (2) the CRD is immovable, so the plan is built backward from it with a buffer; (3) one engineer benefits from small, testable increments, each ending in a working, tested slice; (4) each increment adds a part that can be tested against the decision table and golden test.

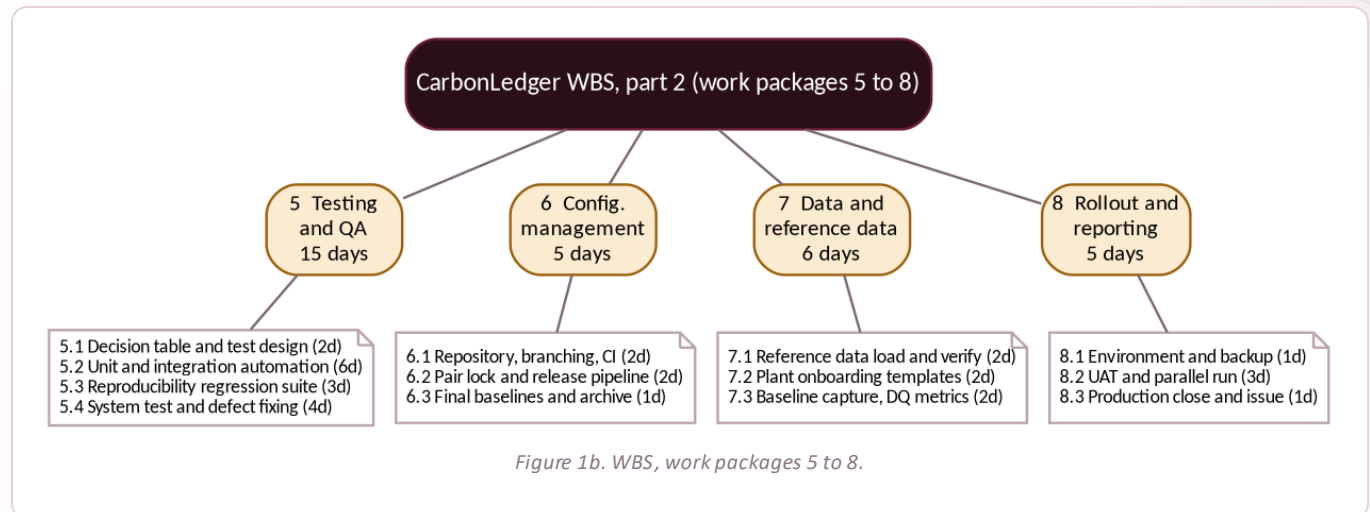
Increment	Content	Exit test
Inc 1	Schema, immutability triggers, reference release loader and sealing	TC-FV-01 to 05
Inc 2	Ingest, tiering, apportioning, calculation, uncertainty	TC-IN, TC-CA, TC-UN, TC-DT rules R1 to R6
Inc 3	Period close, gap-fill, late data, restatement, report builder, canonical JSON	TC-PC, TC-LT, TC-DT rules R7 to R12, TC-RP-02, 03
Inc 4	verify CLI, audit log, roles, interface polish	TC-RP-04 to 08, TC-AU

3 Work Breakdown Structure

The WBS has eight work packages and 27 tasks. Effort is in working days.



3. Work Breakdown Structure



ID	Task	Effort (days)	Weeks (from CRD)
1	Project management	9	
1.1	Planning and tracking	3	W-20 to W-1
1.2	Risk and status reporting	3	W-20 to W-1
1.3	Closure and handover	3	W-1 to W0
2	Requirements	8	
2.1	Elicitation and contract review	3	W-20 to W-19
2.2	SRS writing	4	W-20 to W-19
2.3	SRS review and baseline	1	W-18 to W-18
3	Design	7	
3.1	Temporal data model	3	W-18 to W-17
3.2	Class diagram and module design	2	W-17 to W-16
3.3	Activity diagrams and flows	1	W-18 to W-17
3.4	Design review and baseline	1	W-16 to W-16
4	Build	30	
4.1	Inc 1: schema, immutability, ref loader	6	W-16 to W-14
4.2	Inc 2: ingest, tiering, calc, uncertainty	9	W-13 to W-10
4.3	Inc 3: close, late data, reports	9	W-9 to W-6
4.4	Inc 4: verify, audit, roles, UI	6	W-7 to W-4
5	Testing and QA	15	
5.1	Decision table and test design	2	W-18 to W-16
5.2	Unit and integration automation	6	W-15 to W-5
5.3	Reproducibility regression suite	3	W-11 to W-8
5.4	System test and defect fixing	4	W-5 to W-3
6	Configuration management	5	
6.1	Repository, branching, CI	2	W-19 to W-17
6.2	Pair lock and release pipeline	2	W-14 to W-14
6.3	Final baselines and archive	1	W-3 to W-2
7	Data and reference data	6	

3. Work Breakdown Structure

ID	Task	Effort (days)	Weeks (from CRD)
7.1	Reference data load and verify	2	W-15 to W-15
7.2	Plant onboarding templates	2	W-11 to W-9
7.3	Baseline capture, DQ metrics	2	W-9 to W-5
8	Rollout and reporting	5	
8.1	Environment and backup	1	W-5 to W-3
8.2	UAT and parallel run	3	W-4 to W-3
8.3	Production close and issue	1	W-2 to W-1

Total effort **85 days** against capacity of **100 days**, leaving **15 days (15%) of contingency** that is not allocated to any task.

4

Schedule: Gantt chart aligned to the CRD

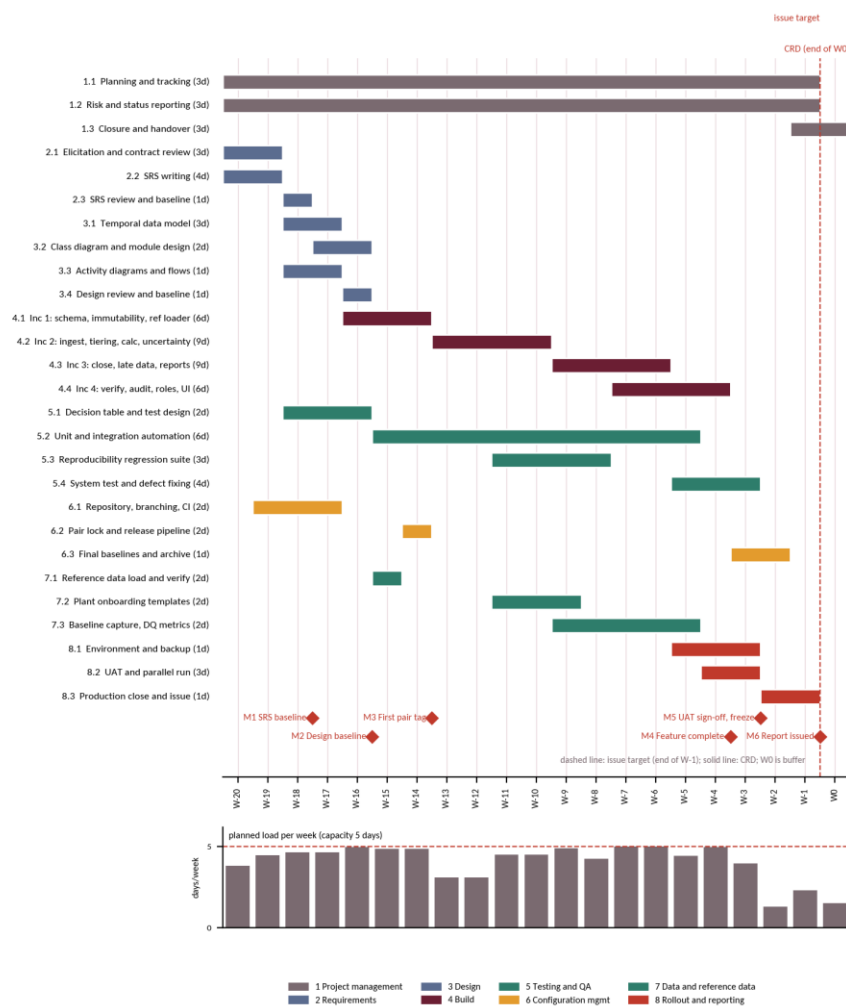


Figure 2. Gantt chart in weeks counted back from the CRD, with planned weekly load. The solid line is the CRD; the dashed line is the target for issuing the report.

4. Schedule: Gantt chart aligned to the CRD

How the plan is aligned to the CRD.

- The report is to be **issued by the end of W-1**, one full week before the CRD (W0), so that W0 is a buffer for customer questions and any correction.
- Period close happens in **W-2**, so there is a further week between close and issue for review and the self-verification step.
- Release freeze and the Pair tag come before close (milestone M5), so the code and reference data used for the issued report are fixed in advance.
- The weekly load never exceeds the 5-day capacity (lower chart); the 15 unallocated days sit mostly in the last six weeks as protection for the date.
- If the CRD moves, every week label shifts by the same amount; only tasks 2.1 and 7.x need re-checking against external dates.

4.1 Milestones

ID	Milestone	Planned	Evidence
M1	SRS and Decision Table baselined	End of W-18	Tag BL-REQ-1.0 ; Sponsor sign-off
M2	Design and CM Plan baselined	End of W-16	Tag BL-DES-1.0
M3	First Pair tagged; golden test passing	End of W-14	Tag CL-<year>.1
M4	Feature complete	End of W-4	Test suite green, 12 rules and 48 combinations exercised
M5	UAT sign-off and release freeze	End of W-3	Release candidate rc-1.0 ; Pair frozen before close
M6	Report issued to customer	End of W-1	Report hash, golden directory added
CRD	Contractual reporting date	End of W0	Contract

4.2 Reporting cycle calendar (working days before the CRD, T = CRD)

This is the calendar the plants and the Reporting Lead follow in the first production cycle, and in every later cycle.

Day	Step	Role
T-25	Data request to plants; declaration requests to suppliers	Plant Manager, Procurement
T-20	First reminder to non-responders	Plant Data Officer
T-15	Second reminder; escalation	Plant Manager
T-14	Administrative data cut-off ; expected items without a record become Open Items	System
T-13 to T-11	Resolve Open Items; Plant Manager confirmations; gap-fill review	Plant Manager, Reporting Lead
T-10	Period close ; report built; self-verify must PASS	Reporting Lead
T-9 to T-6	Review of report and uncertainty disclosure	Reporting Lead, Sponsor
T-5	Issue to customer	Reporting Lead

4. Schedule: Gantt chart aligned to the CRD

Day	Step	Role
T-4 to T0	Buffer; customer queries; late data follows the late-data path	Reporting Lead

5 Resources and roles

Project role	Held by	Main tasks
Project Manager	Sole Engineer	Plan, track, risks, reporting
Business Analyst	Sole Engineer	Requirements, contract review, SRS
Architect and Developer	Sole Engineer	Design pack, build
Tester	Sole Engineer	Decision table, tests, regression suite
Configuration Manager	Sole Engineer	Repository, Pairs, baselines
Sponsor	Manufacturer	Approvals, scope decisions, escalation
Reporting Lead	Manufacturer	Requirements review, UAT, close and issue
Reference Data Steward	Manufacturer	Factor releases and checks
Plant Managers (nine) and Plant Data Officers (nine)	Manufacturer	Data, onboarding, UAT
Customer representative	Customer	Agrees method statement and restatement criterion

Independence note. Because one person holds every project role, independent review is replaced by automation (CI gates, golden tests, determinism checks, the checklist in CL-CMP-001 section 5.3) and by named reviews with the manufacturer's roles at each baseline. This is recorded as risk R-04.

Tools: Git, a pipeline runner, pytest, PostgreSQL, Docker, Graphviz and a document converter for diagrams and documents. Cost is limited to a server and storage; no licences are needed.

6 Risk register

Scoring: Likelihood L and Impact I are each 1 (low), 2 (medium), 3 (high). Score = L x I. **High** = 6 to 9, **Medium** = 3 to 4, **Low** = 1 to 2.

ID	Risk	L	I	Score	Response	Mitigation and contingency	Trigger	Owner (role)
R-01	Supplier non-response: declarations or invoice data not received by the cut-off	3	2	6 High	Mitigate, then accept residual	Requests at T-25; reminders T-20 and T-15; escalation to Procurement; at close the item is gap-filled as ESTIMATED (±25%), listed and disclosed; late declarations handled by rule R10; response tracked as DQ-10; repeat non-responders raised with the Sponsor. Contingency: report with the disclosed higher uncertainty rather than delay.	No reply after second reminder	Plant Manager, Procurement

6. Risk register

ID	Risk	L	I	Score	Response	Mitigation and contingency	Trigger	Owner (role)
R-02	Authority publishes factors late, or corrects them retroactively	2	3	6 High	Mitigate	Confirm publication timetable at task 2.1; load release by W-14; if a factor is missing at close, close is blocked, so agree an interim approach with the customer in writing; corrections become a new release and, if needed, a restatement.	Edition not published by W-14	Reference Data Steward
R-03	CRD moves earlier, or the contract changes period length or format	2	3	6 High	Mitigate	Plan counted back from CRD; one-week buffer W0; 15 contingency days; change control; periods are configuration.	Written notice from customer	Project Manager
R-04	Sole engineer unavailable (illness, examinations)	2	3	6 High	Mitigate	Documents and procedures kept current; automation instead of memory; offline bundle of repository; 15-day buffer; scope reduction by MoSCoW (section 8.3).	Absence of more than 3 days	Project Manager
R-05	Scope creep (market-based Scope 2, factor uncertainty, automatic bill reading)	2	2	4 Medium	Avoid	Out-of-scope list in SRS 1.2; change control; deferred to a later version.	Any new request after M1	Project Manager
R-06	Plant data worse than expected: high estimated share, inconsistent formats	3	2	6 High	Mitigate	Entry templates and CSV validator; onboarding W-11 to W-9; Cycle 0 baseline; Data Quality Plan targets; disclosure rather than concealment.	Estimated share high at baseline	Plant Manager
R-07	Large late invoices change totals after close	2	2	4 Medium	Mitigate	Late-data rules, adjustment lines, restatement criterion agreed with the customer by W-10; watchlist.	Adjustments reach restatement criterion	Reporting Lead
R-08	Metering delayed (procurement, installation lead time)	2	2	4 Medium	Accept and plan	Cycle 1 relies on invoiced conversion (74% of the benefit); metering plan from Cycle 2; targets revisable once after Cycle 0.	No meter by planned cycle	Plant Manager
R-09	Non-determinism enters (library update, ordering, locale) and breaks byte equality	2	3	6 High	Prevent	Decimal arithmetic; hash-locked dependencies; pinned image; TC-RP-03 and golden suite in CI on every commit.	Golden test fails without a code change	Sole Engineer (Tester)
R-10	Error in reference data load (wrong unit, wrong year, wrong row)	2	3	6 High	Prevent	Row-by-row diff against the publication (CL-CMP-001 5.4); unit conversion tests; manifest hash; review by Reporting Lead.	Diff shows unexplained change	Reference Data Steward
R-11	Customer disputes method (RSS, tier percentages, gap-fill, restatement rule)	2	3	6 High	Mitigate	Share SRS section 3.4 to 3.6 at M1; disclose both RSS and worst-case bound; method statement in every report.	Objection at review	Sponsor
R-12	Data or evidence lost or corrupted	1	3	3 Medium	Mitigate	Daily backup; restore test each quarter; hashes; offline bundle.	Failed restore test	Sole Engineer
R-13	Unauthorised change to data or factors	1	3	3 Medium	Prevent	Roles, database rights, immutability triggers, audit log, physical audit.	Audit mismatch	Sole Engineer
R-14	Performance below target	1	1	1 Low	Accept	TC-PF-01 run early in Inc 2.	Recompute takes longer than the target	Sole Engineer
R-15	Low adoption by plant staff	2	2	4 Medium	Mitigate	Single-form entry; training at onboarding; Plant Managers own metrics.	High after-cut-off share	Plant Manager
R-16	Synthetic test factors reach production	1	3	3 Medium	Prevent	Release ids with prefix RR-T are blocked by a production profile check and a CI test.	Check fails	Sole Engineer
R-17	Unfamiliar database features (range types, exclusion constraints) delay Inc 1	2	2	4 Medium	Mitigate	One-day spike in W-16; fall back to trigger-based overlap check.	Spike not working by end of day 2	Sole Engineer
R-18	Effort underestimated	2	2	4 Medium	Mitigate	15-day contingency; weekly effort tracking; early warning threshold in section 8.	Burn above plan two weeks running	Project Manager

Risk matrix (count of risks per cell):

6. Risk register

Likelihood \ Impact	1 Low	2 Medium	3 High
3 High	0	R-01, R-06	0
2 Medium	0	R-05, R-07, R-08, R-15, R-17, R-18	R-02, R-03, R-04, R-09, R-10, R-11
1 Low	R-14	0	R-12, R-13, R-16

Risks are reviewed weekly (task 1.2). A risk moves to Medium or Low only when its mitigation is evidenced, not merely planned.

7 Quality plan

Area	Standard	Check
Requirements	Each requirement testable, with an ID and a rationale	Review at M1; traceability matrix in CL-DTT-001
Design	Every requirement group maps to a module and table	Review at M2
Code	Typed Python, formatted and linted in CI; calculation modules are pure functions	CI on every commit
Tests	All 12 rules and 48 combinations; golden and mutation checks	Entry and exit criteria in CL-DTT-001 section 10
Documents	Controlled, versioned, cross-referenced	Baseline audit (CL-CMP-001 section 8)
Definition of done (for any task)	Tests written and passing; documented; merged to <code>main</code> ; baseline tag where applicable	Checked at merge

8 Monitoring and control

8.1 Tracking

Weekly, the Sole Engineer records effort spent against the plan in the Gantt load chart, milestones met, risks changed and decisions needed, and sends a one-page status to the Sponsor every two weeks.

8.2 Early-warning thresholds

Indicator	Threshold	Action
Cumulative effort ahead of plan	More than 5 of the 15 buffer days used before W-10	Apply scope reduction (8.3)
A milestone slips	More than one week	Re-plan remaining work; inform Sponsor; review risks R-03 and R-18
Golden test fails	Any time	Stop feature work until fixed (R-09)

8. Monitoring and control

Indicator	Threshold	Action
Cut-off approaching with Open Items	At T-14	Plan gap-fill with Reporting Lead

8.3 Scope reduction order (MoSCoW)

Priority	Items
Must	Factor versioning, calculation, uncertainty disclosure, period close, late-data rules, canonical report, verify, golden test, audit log, roles
Should	Plant onboarding templates, CSV import, PDF and CSV renderings polish, per-plant scorecard
Could	Interface polish, drill-down convenience, extra charts

If the plan must shrink, "Could" items go first, then "Should"; "Must" items are never traded for date.

8.4 Change management

Changes follow CL-CMP-001 section 5. Any change that affects scope or the CRD plan is decided by the Sponsor and recorded with its effect on effort, buffer and risks.

9 Communication plan

Audience (role)	Content	Frequency	Channel
Sponsor	Status, risks, decisions needed	Every two weeks and at each milestone	One-page report, short meeting
Reporting Lead	Requirements review, UAT, close planning	Weekly from W-11	Demo and checklist
Plant Managers and Data Officers	Onboarding, data calendar, scorecard	At W-11 to W-9, then each cycle	Session and written guide
Customer representative	Method statement, restatement criterion, report	At M1, before issue, at issue	Written
Reviewer (Faculty)	Documents and milestones	At baselines	Documents

10 Acceptance

The project is accepted when: the first production period is closed and the report issued by end of W-1; **verify** returns PASS on it and its golden directory exists; all exit criteria in CL-DTT-001 section 10 are met; the Data Quality Plan baseline has been captured; and the six documents are baselined at v1.0.